

# Md Showkat Hossain Chy

Curriculum vitae

# Md Showkat Hossain Chy

Seven years building systems where correctness under concurrency, failure, and scale is the hard part: firmware under fixed memory and power budgets, distributed services that reconcile out-of-order events, and machine learning over graphs with millions of nodes.

Seattle, Washington, USA — U.S. Permanent Resident — mdshowkathossainchy@gmail.com

## Research record

12

peer-reviewed papers

148

citations

8

h-index

Venues include IEEE/ACM ASE, ECSA, CLOSER, Journal of Systems and Software, Cluster Computing, Electronics, Telecom, SeGAH, and ICSEC.

## Education

Jan 2022 - Dec 2023

### Baylor University

M.S. in computer science, Waco, Texas

Concentration in machine learning for software systems, microservice architecture, and distributed data processing.

Jan 2013 - May 2018

### Daffodil International University

B.Sc. in computer science and engineering, Dhaka, Bangladesh

## Experience

May 2026 - present

### Founder and engineer

Stealth AI startup, Seattle, Washington

Building an agentic AI platform that monitors municipal records, council agendas, zoning dockets, and permit filings across U.S. cities, and converts them into early business signals for commercial real estate and multi-site operators.

Jan 2025 - Apr 2026

## Software development engineer, AWS infrastructure supply chain

Amazon Web Services, Seattle, Washington

Modelled millions of hardware components, assemblies, and bills of materials as a graph in Amazon Neptune and DynamoDB, supporting change-impact traversal across products, sites, and programs.

Held the system of record to a 24-hour consistency commitment with an event-driven reconciliation service on EventBridge, SQS, AWS Batch, and DynamoDB that detected and safely replayed dropped, duplicated, and out-of-order updates.

Built the platform end to end: a Java and Spring Boot rules engine evaluating site, region, and program policy from declarative configuration, through a GraphQL, React, and TypeScript interface. 60% gain in hardware qualification efficiency, with new categories onboarded across 100+ manufacturing sites without per-class code.

Integrated 15+ cross-region systems behind versioned REST APIs with role-based access control and per-caller rate limits.

Raised integration test coverage from 60% to 90% with Cypress, and instrumented the platform with CloudWatch, real user monitoring, CodeGuru profiling, and Lambda-based certificate monitoring.

Mar - Dec 2024

## Research associate II, software engineering

CloudHubs Lab, University of Arizona, Tucson, Arizona

Led the lab's graph-learning line: graph neural networks and variational autoencoders with fuzzy c-means clustering over large static-analysis dependency graphs, recovering latent structure automatically. Published at IEEE/ACM ASE 2024 and in Cluster Computing.

Designed and released a multi-variant benchmark suite giving controlled, reproducible subjects for architecture research.

Characterised how changes propagate across service boundaries.

Mentored 15 graduate and undergraduate students in experimental design and empirical evaluation.

Mar - Dec 2024

## Full-stack developer

ICPC Foundation, University of Arizona, Tucson, Arizona

Built contest-management workflows in Spring Boot, React, PostgreSQL, and REST covering registration, scheduling, and logistics for 50,000+ students annually across 50+ regions.

Designed a multi-tenant, configuration-driven platform letting regional managers publish events independently. 45% reduction in administrative time, \$12K saved annually.

Ran static analysis with SonarQube, resolving 1,600+ defects and security issues.

May - Aug 2023

## Software engineer intern

Tyson Foods, Springdale, Arkansas

Delivered Unreal Engine virtual-reality quality-inspection training for 200+ manufacturing personnel, worth \$233K in annual savings.

Automated 3D asset generation, cutting a two-week manual process to 10 minutes on an AWS EC2 and Perforce pipeline.

Jan 2022 - Dec 2023

## Graduate research and teaching assistant

Baylor University, Waco, Texas

Built distributed MapReduce and PySpark pipelines on Amazon EMR over 10M+ blockchain transactions, extracting opcode and control-flow features from Solidity bytecode and training anomaly-detection models. 65% improvement in analysis efficiency.

Ran a controlled empirical evaluation of four JVM message queues (Kafka, Artemis, Pulsar, RocketMQ), measuring latency, message ordering, and delivery reliability under concurrency.

Implemented a transaction manager in Python enforcing ACID guarantees through strict two-phase locking and write-ahead logging, with crash recovery under concurrent workloads.

Taught software engineering (Java) and C++ programming: 125 students across five sections.

Jul 2018 - Dec 2021

## Software engineer, wearable platform

Samsung Research, Dhaka, Bangladesh

Developed FreeRTOS firmware and device drivers in C and C++ over SPI, UART, and I2C for heart-rate, stress, and sleep tracking across 90+ exercise modes. Profiled and debugged races and timing defects at the microcontroller level with JTAG and Trace32 under fixed memory and power budgets.

Designed the Bluetooth Low Energy and GATT control and data planes — service discovery, characteristics, message framing, connection stability — sustaining real-time sensor streaming on a 15 ms cadence.

Founding engineer on phone-to-wearable connectivity shipped to 2M+ devices with no known production defects. Authored the Samsung Accessory Protocol libraries in C and C++ for Android and iOS.

Built rollback-safe over-the-air update workflows with recovery paths for devices that could not be physically retrieved: 95% test coverage, 99% rollout success, zero bricked devices.

Jan - Jun 2018

## Software engineer

Divine IT Limited, Dhaka, Bangladesh

Built Django, DRF, and PostgreSQL ERP modules across payroll, HR, inventory, sales, and procurement for 5,000+ daily users, improving throughput by 30%.

## Technical range

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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages              | C, C++, Java, Python, TypeScript, JavaScript, SQL, Bash                                                                                                                                                                                         |
| Embedded and low-level | FreeRTOS, device drivers over SPI, UART, and I2C, Bluetooth Low Energy and GATT, JTAG and Trace32 profiling, memory- and power-constrained optimisation, rollback-safe over-the-air updates                                                     |
| Distributed and cloud  | Event-driven architecture, MapReduce, PySpark, Amazon EMR, Kafka, Pulsar, Artemis, RocketMQ, DynamoDB, Amazon Neptune, PostgreSQL, Redis, Docker, Kubernetes, and AWS (Lambda, EventBridge, SQS, SNS, Batch, EC2, S3, IAM, CloudFormation, CDK) |
| Machine learning       | Graph neural networks, variational autoencoders, representation learning over graphs, clustering, anomaly detection, PyTorch, TensorFlow                                                                                                        |
| Engineering practice   | JUnit, Cypress, integration and regression testing, coverage analysis, CI/CD, SonarQube, static analysis, Spring Boot, GraphQL, REST, Git, Perforce                                                                                             |

## Publications

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### Journal articles

#### Multivocal Study on Microservice Dependencies

A. S. Abdelfattah, T. Cerny, **M. S. H. Chy**, M. A. Uddin, S. Perry, C. Brown, J. Yero, K. Sooksatra

**JSS** 2025 volume 222, article 112334

doi 10.1016/j.jss.2025.112334

#### Benchmarking Micro2Micro Transformation: An Approach with Graph Neural Networks and Variational Autoencoders

**M. S. H. Chy**, K. Sooksatra, J. Yero, T. Cerny

**Cluster Computing** 2024 volume 27, number 4, pages 4171–4185

doi 10.1007/s10586-024-04526-z

#### Comparative Evaluation of Java Virtual Machine-Based Message Queue Services: A Study on Kafka, Artemis, Pulsar, and RocketMQ

**M. S. H. Chy**, M. A. R. Arju, S. M. Tella, T. Cerny

**Electronics** 2023 volume 12, article 4792

doi 10.3390/electronics12234792

#### Benchmarking Message Queues

R. Maharjan, **M. S. H. Chy**, M. A. R. Arju, T. Cerny

**Telecom** 2023 volume 4, number 2, pages 298–312

doi 10.3390/telecom4020018

### Conference papers

#### Using Static Analysis to Aid Monolith to Microservice System Transformation: Tuning Fuzzy c-Means in a VAE-Based GNN Approach

K. Sooksatra, **M. S. H. Chy**, M. A. R. Arju, T. Cerny, P. Rivas

**IEEE/ACM ASE** 2024

doi 10.1145/3691621.3694933

#### A Multi-variant Benchmark for Microservice Systems in Software Engineering Research

T. Cerny, **M. S. H. Chy**, M. A. R. Arju, K. Sooksatra, A. S. Abdelfattah, V. Lenarduzzi

**ECSA** 2024 pages 21–29

doi 10.1007/978-3-031-71246-3\_3

#### On Maintainability and Microservice Dependencies: How Do Changes Propagate?

T. Cerny, **M. S. H. Chy**, A. S. Abdelfattah, J. Soldani, J. Bogner

**CLOSER** 2024 volume 1, pages 277–286

doi 10.5220/0012725200003711

#### On Dependencies in Microservices: Dependency Management and Maintainability

T. Cerny, **M. S. H. Chy**, M. A. Uddin, A. S. Abdelfattah, J. Soldani, J. Bogner

**Springer CCIS** 2024 pages 29–52

doi 10.1007/978-3-032-17286-0\_2

#### Evolution and Anti-patterns Visualized: MicroProspect in Microservice Architecture

L. Adams, A. Elsayed, **M. S. H. Chy**, S. Perry, P. Harris, T. Cerny, D. Amoroso d'Aragona, D. Taibi

**ECSA** 2023 pages 309–325

doi 10.1007/978-3-031-66326-0\_19

## A Systematic Review on Software Design Patterns in Today's Perspective

M. Rahman, **M. S. H. Chy**, S. Saha

IEEE SeGAH

2023 Athens, Greece, pages 1–8

## Bless: An Interactive Application to Show Gratitude towards Doctors

M. Rahman, M. S. Rahaman, **M. S. H. Chy**

IEEE SeGAH

2023 Athens, Greece, pages 1–7

## Exploring Chromatin Interaction Between Two Human Cell Types and Different Normalization Techniques for Hi-C Data

M. Rahman, **M. S. H. Chy**, A. Rahman, S. Ahmed, T. Sharmin, M. S. Rahaman

ICSEC

2023 Thailand, pages 53–59

## Academic service and peer review

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Program chair, 32nd International Conference on Information Systems Development, 2024.

Program chair, 18th IEEE International Conference on Service-Oriented System Engineering, 2024.

Reviewer, Springer Cluster Computing, 2024.

Member, IEEE, IEEE Computer Society, and IEEE Young Professionals, 2024 to present.

## Teaching and mentoring

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Teaching assistant, software engineering (Java) and C++ programming, Baylor University — 125 students across five sections, 2022–2023.

Research mentor, CloudHubs Lab, University of Arizona — 15 graduate and undergraduate students in software-architecture research and empirical methods, 2024.

Capstone supervisor, Baylor University, from requirements through final delivery.

Engineering mentor, Samsung Research — four junior engineers in debugging, test-driven development, and code review, 2018–2021.

Contest trainer, judge, and problem setter, Daffodil International University, 2015–2017.

## Competitive programming and awards

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ACM ICPC contestant, 2013–2016 — four Dhaka Site Regionals, six national contests, around 1,200 problems solved.

First runner-up, Inter-University Programming Contest, 2014. Champion, DIIT Code Fest, 2015.

Chief Technology Officer Award, Samsung Electronics, for the Galaxy Fit project, 2019.

## Languages

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English — IELTS overall band 7.5, listening 8.5, reading 8.0.

Bengali — native proficiency.